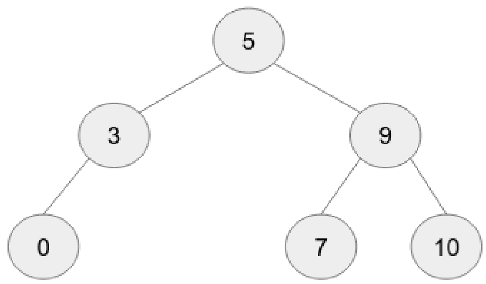
January 2024, CSE 106
Online on Binary Search Tree (A1/A2)
Time: 40 Minutes

Problem Specification: Your task is to implement the following function.

```
Node* lca(Node* node, int a, int b)
```

Here, `node` is the root node of a BST. Integers, a and b, are guaranteed to exist in the tree. The function returns the tree node that is the lowest common ancestor of a and b. The **Lowest Common Ancestor (or LCA)** is the lowest node in the BST such that both a and b are its **descendants** in the BST. For example, in the following BST, the LCA of 7 and 10 is the node with value 9. Similarly, the LCA of 0 and 9 is the root node.



Input/Output Format: The input-output format is similar to the one described in *Assignment 4: Binary Search Tree*. In addition, the LCA query will start with L followed by two integers. A modified `main.cpp` is uploaded to evaluate your solution.

Assuming that the above tree is already built, sample inputs and outputs are as follows.

Sample Inputs	Sample Outputs
L 7 10	9
L 0 3	3

Hint: From any `node`, using the properties of a binary search tree, you can determine whether both a and b are in the same sub-tree or different sub-tree.

Submission Guidelines: Save your `.c` file as `2306<3-digit-Student-ID>.c` and upload it on Moodle in the corresponding submission link.